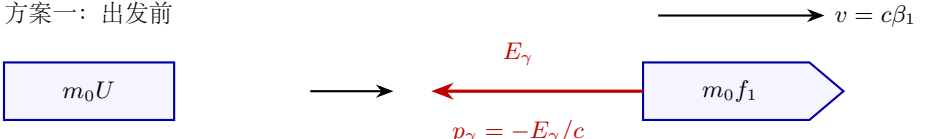
方案一：出发前



$v = 0, \quad p_{\text{initial}} = 0$

初始总能量 m_0Uc^2

$E_{\text{ship}} = \gamma_1 m_0 f_1 c^2$

$p_{\text{ship}} = \gamma_1 m_0 f_1 \beta_1 c$

方案二：船与待收集部分

$v_{\text{in}} = 0$

两者初始动量均为零

总能量 $m_0(A + X)c^2$

$E_{\text{ship}} = \gamma_2 m_0 f_2 c^2$

$p_{\text{ship}} = \gamma_2 m_0 f_2 \beta_2 c$